数理分析方法

函数、对应与可数集

Lecture 103

原课件逐页保留 · 零基础中文讲解 · 证明训练

材料与编写说明

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

怎样阅读

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

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Contents

零基础读证明：随手查阅 3
1 原课件第 1 页：函数与对应的入口 4
2 原课件第 2 页：本讲的位置 5
3 原课件第 3 页：函数是什么 6
4 原课件第 4 页：函数的图 7
5 原课件第 5 页：像与满射 8
6 原课件第 6 页：像保持并集 9
7 原课件第 7 页：指标写法与反例 10
8 原课件第 8 页：逆像不用反函数 11
9 原课件第 9 页：逆像与逻辑 12
10 原课件第 10 页：逆像公式的另一写法 13
11 原课件第 11 页：单射与双射的区别 14
12 原课件第 12 页：延拓与限制 15
13 原课件第 13 页：坐标投影 16
14 原课件第 14 页：对应与经济选择 17
15 原课件第 15 页：有效定义域 18
16 原课件第 16 页：集合的对应像 19
17 原课件第 17 页：对应像的运算 20
18 原课件第 18 页：补集结论的证明 21
19 原课件第 19 页：对应的图 22
20 原课件第 20 页：二元运算 23
21 原课件第 21 页：交换律与结合律 24
22 原课件第 22 页：分配律 25
23 原课件第 23 页：对应逐点运算 26
24 原课件第 24 页：对应的复合 27
25 原课件第 25 页：常对应与恒等对应 28
26 原课件第 26 页：复合结合与可达性 29
27 原课件第 27 页：半单值及单射 30
28 原课件第 28 页：逆对应 31
29 原课件第 29 页：下逆像 32
30 原课件第 30 页：上逆像 33
31 原课件第 31 页：补集对偶 34
32 原课件第 32 页：并集的上下逆像 35

33 原课件第 33 页：数列是函数 36
34 原课件第 34 页：可数与有限乘积 37
35 原课件第 35 页：Cantor 对角证明 38

零基础读证明：随手查阅

对象的层次。 $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 A 的每个元素都在 B 中。$\{x\}$ 是只含 x 的集合，不是 x 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 N 个坐标的有序列表。

量词。 $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

怎样否定。 否定“所有 x 满足 P”得到“存在 x 不满足 P”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 P 真且 Q 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

四种基本证明。 (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

数列、子列和收敛。 数列 x_n 是第 n 个对象；子列 x_{n_k} 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

距离的关键估计。 $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 ε；分成 m 段时可各分配 ε/m。这里的 ε 不是固定的小数，而是任意正精度。

做证明题的五行草稿。 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

1 原课件第 1 页：函数与对应的入口

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

September 10, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本讲把“唯一输出”的函数推广为“输出集合”的对应。先学集合元素与子集的区别：$y \in \Phi(x)$，而不是把集合 $\Phi(x)$ 当成数。

2 原课件第 2 页：本讲的位置

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

集合运算用于处理像与逆像，二元运算用于建立实数代数，后面的连续性则通过逆像描述。先把符号翻译为中文句子，再证明集合等式。

3 原课件第 3 页：函数是什么

Functions

Let S and T be two sets; if with each element $x \in S$ is associated one and only one element $y \in T$, a function f from S to T is defined. f is also called a transformation of S into T. x is the variable, y is the image of x by f, or the transform of x by f, or the value of f at x, and one writes $y = f(x)$ (read y equals f of x), or $x \to f(x)$.

Example: Let A be the set of actions available to a certain agent. If the choice of a in A determines the amount of money (a positive integral number of cents) this agent receives, a function from A to $\mathbb{N}$ is defined.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$f : S \to T$ 要求每个 $x \in S$ 恰有一个输出。定义域是 S，陪域是 T。例如 $f(x) = x^2$ 从实数到实数是函数，虽然不同输入可能有相同输出。

4 原课件第 4 页：函数的图

Functions

Consider in $S \times T$ the set of elements (x, y) for which $y = f(x)$. This subset of $S \times T$ is called the graph of the function f.

$$G(f) = \{(x, y) \in S \times T \mid y = f(x)\}.$$

Example: The graph of a function f from S to T is visualized as a set of points of the rectangle such that the vertical through an arbitrary point x of S intersects it at exactly one point.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

图是所有 $(x, f(x))$ 的集合。证明某集合是函数图，要检查每个输入存在且仅有一个对应点；“画得像曲线”不构成证明。

5 原课件第 5 页：像与满射

Functions

Let X be a subset of S; take the image $y = f(x)$ of each $x \in X$. The set of the images so obtained is a subset of T called the image of X and denoted $f(X)$.

$$f(X) = \{f(x) \in T \mid x \in X\}.$$

If $f(S)$ consists of one element of T, in other words, if all the elements of S have the same image in T, the function f is said to be constant. If $f(S) = T$, in other words, if each element of T is the image of some element of S, f is said to be a function from S onto T, i.e., function $f : S \to T$ is a surjective (onto) mapping.

For any $y \in T$, there exists $x \in S$ such that $f(x) = y$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$y \in f(A)$ 等价于存在 $x \in A$ 使 $f(x) = y$。满射是 $f(S) = T$；改变陪域可改变满射性质，例如平方函数到 $\mathbb{R}$ 不满，到 $[0, \infty)$ 满。

6 原课件第 6 页：像保持并集

Functions

Theorem 1. If f is a function from a set S to a set T, and if $\mathscr{X}$ is a collection of subsets of S, then

(1) $f\left(\bigcup_{X\in\mathscr{X}} X\right) = \bigcup_{X\in\mathscr{X}} f(X)$.

(2) $f\left(\bigcap_{X\in\mathscr{X}} X\right) \subseteq \bigcap_{X\in\mathscr{X}} f(X)$.

Remark: There exist a function f from a set S to a set T, and $\mathscr{X}$ is a collection of subsets of S, such that then $f\left(\bigcap_{X\in\mathscr{X}} X\right) \neq \bigcap_{X\in\mathscr{X}} f(X)$.

Example: Let $A = \{1,2\}$, $f(1) = f(2) = 0$, then $f(\{1\}) \cap f(\{2\}) = \{0\}$ while $\{1\} \cap \{2\} = \varnothing$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$y \in f(\bigcup A_i) \iff \exists i \exists x \in A_i : f(x) = y \iff y \in \bigcup f(A_i)$。交集只得包含：同一个 x 属于所有 A_i 才可直接取像；反向可能由不同原像产生同一个 y。

7 原课件第 7 页：指标写法与反例

Functions

Theorem 1. If f is a function from a set S to a set T, and if $\{X_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of S, then

(1) $f\left(\bigcup_{\lambda\in\Lambda} X_\lambda\right) = \bigcup_{\lambda\in\Lambda} f(X_\lambda)$.

(2) $f\left(\bigcap_{\lambda\in\Lambda} X_\lambda\right) \subseteq \bigcap_{\lambda\in\Lambda} f(X_\lambda)$.

Remark: There exist a function f from a set S to a set T, and $\{X_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of S, such that then $f\left(\bigcap_{\lambda\in\Lambda} X_\lambda\right) \neq \bigcap_{\lambda\in\Lambda} f(X_\lambda)$.

Example: Let $A = \{1, 2\}$, $f(1) = f(2) = 0$, then $f(\{1\}) \cap f(\{2\}) = \{0\}$ while $\{1\} \cap \{2\} = \varnothing$.

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本页把上页集合族改为指标族，证明不变。常值函数 $f(1) = f(2) = 0$ 给出 $f(\{1\} \cap \{2\}) = \varnothing$ 而像的交为 $\{0\}$。反例必须把左右两边都算出。

8 原课件第 8 页：逆像不用反函数

Functions

Conversely, let Y be a subset of T; the set of $x \in S$ which have their images in Y is a subset of S called the inverse image of Y and denoted $f^{-1}(Y)$.

$$f^{-1}(Y) = \{x \in S \mid f(x) \in Y\}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$f^{-1}(B) = \{x : f(x) \in B\}$ 对任何函数都定义。例如平方函数下 $\{1\}$ 的逆像是 $\{-1, 1\}$。逆像是集合，不要求 f 可逆。

9 原课件第 9 页：逆像与逻辑

Functions

Theorem 2. If f is a function from a set S to a set T, and if $\mathscr{Y}$ is a collection of subsets of T, then It is easy to prove that:

(1) $f^{-1}\left(\bigcup_{Y\in\mathscr{Y}} Y\right) = \bigcup_{Y\in\mathscr{Y}} f^{-1}(Y)$.

(2) $f^{-1}\left(\bigcap_{Y\in\mathscr{Y}} Y\right) = \bigcap_{Y\in\mathscr{Y}} f^{-1}(Y)$.

(3) $f^{-1}(Y \setminus Y') = f^{-1}(Y) \setminus f^{-1}(Y')$ for $Y' \subseteq Y \subseteq T$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$x \in f^{-1}(\bigcap B_i) \iff \forall i, f(x) \in B_i \iff x \in \bigcap f^{-1}(B_i)$。把全称改为存在即证并集公式。差集公式来自“属于 B 且不属于 B'”。

10 原课件第 10 页：逆像公式的另一写法

Functions

Theorem 2. If f is a function from a set S to a set T, and if $\{Y_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of T, then It is easy to prove that:

(1) $f^{-1}\left(\bigcup_{\lambda\in\Lambda} Y_\lambda\right) = \bigcup_{\lambda\in\Lambda} f^{-1}(Y_\lambda)$.

(2) $f^{-1}\left(\bigcap_{\lambda\in\Lambda} Y_\lambda\right) = \bigcap_{\lambda\in\Lambda} f^{-1}(Y_\lambda)$.

(3) $f^{-1}(Y \setminus Y') = f^{-1}(Y) \setminus f^{-1}(Y')$ for $Y' \subseteq Y \subseteq T$.

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指标族与集合族表达同一事实，重写上页的三个等价链即可。补集公式是 $f^{-1}(T\setminus B) = S\setminus f^{-1}(B)$，注意左右补集的全集不同。

11 原课件第 11 页：单射与双射的区别

Functions

A $y \in T$ may be the image of several, or of one, or of no element of S. When for each $y \in T$ the set $f^{-1}(y)$ consists of exactly one element, in other words, when each $y \in T$ is associated with one and only one element of S, f is said to establish a one-to-one correspondence between S and T, i.e., function $f : S \to T$ is an injective (one-to-one) mapping.

For any $x_1 \in S$ and $x_2 \in S$, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

For any $x_1 \in S$ and $x_2 \in S$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

纠错：每个 $y \in T$ 恰有一个原像表达双射，不只是单射。单射仅要求 $f(x) = f(x') \Rightarrow x = x'$；陪域中的某些点可以没有原像。双射同时满足单射和满射。

12 原课件第 12 页：延拓与限制



Let f be a function from a set F to a set T, and let G be a set containing F. A function g from G to T is said to be an extension of f to G if one has $f(x) = g(x)$, for every x in F.

Let f be a function from a set F to a set T, and let G be a set contained in F. A function g from G to T is said to be an restriction of f to G if one has $f(x) = g(x)$, for every x in G.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

限制缩小定义域但保持原来输出，延拓扩大定义域并在旧定义域与原函数一致。例：实数平方函数限制到非负实数后成为双射。检查延拓时应逐个核对旧输入，而不是只比较公式名称。

13 原课件第 13 页：坐标投影

Functions

Consider N sets $S_1, \cdots, S_n, \cdots, S_N$ and their product $\prod_{n=1}^{N} S_n$. The function which associates with the generic element (x_n) of $\prod_{n=1}^{N} S_n$, its nth coordinate x_n in S_n is called the nth projection (or the projection on S_n). The image of an element (resp. of a set) by a projection function is called the projection of that element (resp. of that set).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\pi_i(x_1, \ldots, x_N) = x_i$。投影会丢失其他坐标，一般不单射。以后证明向量收敛可逐坐标进行，因为坐标数是有限的。

14 原课件第 14 页：对应与经济选择

Correspondences

If with each element x in a given set S is associated a non-empty subset Y of a given set T, a function φ, from S to the set of subsets of T is defined. It is sometimes preferable to consider φ as a correspondence from S to T. One writes $Y = \varphi(x)$.

Example: (1) Let f be a function from a set S onto a set T; for every $y \in T$, $f^{-1}(y)$ is a non-empty subset of S, hence f^{-1} is a correspondence from T to S. (2) Let A be the set of a priori available actions of a social agent. Suppose that his environment is completely specified by an element e of a set E. His environment restricts his freedom of action, i.e., the element e determines the subset of A to which his choice is actually restricted. A correspondence from E to A is thus introduced.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\Phi(x) \subseteq T$ 可以含多个元素，例如预算下所有可行消费。课件有时默认值非空，有时允许空值；每道题应说明采用哪种约定。

15 原课件第 15 页：有效定义域

Correspondences

Let S and T be two sets. If with each element x of S we associate a subset $\varphi(x)$ of T, we say that the correspondence $x \longrightarrow \varphi(x)$ is a mapping of S into T; the set $\varphi(x)$ is called the image of x under the mapping φ. The set $S^* = \{x \in S \mid \varphi(x) \neq \varnothing\}$ is called the domain (or set of definition) of φ and $T^* = \bigcup_{x \in S} \varphi(x)$ is called the range (or set of values) of φ; we also say that φ is defined on S^* and that it is a mapping of S^* onto T^*, i.e., correspondence $\varphi : S^* \to T^*$ is a surjective (onto) mapping.

If the mapping φ of S into T is such that the set $\varphi(x)$ always consists of a single element, we say that φ is a single-valued function or a single-valued mapping of S into T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$S^* = \{x : \Phi(x) \neq \varnothing\}$，值域为所有输出的并。单值且处处非空时可以视为函数。不要把“对应的值域”等同于一开始指定的陪域。

16 原课件第 16 页：集合的对应像

Correspondences

A mapping is called semi-single-valued if

$$\varphi(x) \cap \varphi\left(x'\right) \neq \varnothing \Longrightarrow \varphi(x) = \varphi\left(x'\right).$$

Clearly a single-valued mapping is also semi-single-valued. A mapping is called injective (one-to-one) if

$$x \neq x' \Longrightarrow \varphi(x) \cap \varphi\left(x'\right) = \varnothing.$$

An injective mapping is evidently semi-single-valued.

If φ is a mapping of S into T and A is a non-empty subset of S, we write

$$\varphi(A) = \bigcup_{x \in A} \varphi(x).$$

If $A = \varnothing$, we write $\varphi(\varnothing) = \varnothing$. The set $\varphi(A)$ is called the image of A under the mapping φ; if we compare x to a source of light, making a shadow $\varphi(x)$ on a screen, then $\varphi(A)$ is the shadow produced by a set A of sources of light.

If $\mathscr{A} = \{A_\lambda \mid \lambda \in \Lambda\}$ is a family of sets, we write

$$\varphi(\mathscr{A}) = \{\varphi\left(A_\lambda\right) \mid \lambda \in \Lambda\}$$

and call $\varphi(\mathscr{A})$ the image of $\mathscr{A}$ under the mapping φ.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\Phi(A) = \bigcup_{x \in A} \Phi(x)$，即从 A 中至少一个输入可以达到的输出。若 A 为空，像按约定为空。集合族的像则是由这些像组成的新集合族，层次不同。

17 原课件第 17 页：对应像的运算

Correspondences

Theorem 3. $A \subseteq A'$ implies that $\varphi(A) \subseteq \varphi(A')$.

Theorem 4. $\varphi\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) = \bigcup_{\lambda\in\Lambda} \varphi(A_\lambda)$.

Theorem 5. $\varphi\left(\bigcap_{\lambda\in\Lambda} A_\lambda\right) \subseteq \bigcap_{\lambda\in\Lambda} \varphi(A_\lambda)$.

Theorem 6. If $A \subseteq S$ and if φ is a mapping of S with range T, then

$$\mathbf{C}(\varphi(A)) \subseteq \varphi(\mathbf{C}A).$$

If, further, φ is injective, then

$$\mathbf{C}(\varphi(A)) = \varphi(\mathbf{C}A).$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

单调性：$A \subseteq B$ 时任一 $y \in \Phi(A)$ 有见证 $x \in A \subseteq B$，故属于 $\Phi(B)$。并集交换存在量词得等式；交集只有包含。补集结论必须相对实际值域，否则未达到的输出没有原像可选。

18 原课件第 18 页：补集结论的证明

Correspondences

Proof of Theorem 6. $y \in \mathbf{C}(\varphi(A))$, $y \notin \varphi(A) = \bigcup_{x \in A} \varphi(x)$, there isn't $x \in A$ such that $y \in \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$, and hence $y \in \varphi(\mathbf{C}A)$.

Conversely, $y \in \varphi(\mathbf{C}A) = \bigcup_{x \in \mathbf{C}A} \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$. If φ is injective, then $y \notin \varphi(x)$ for $x \in A$, and $y \in \varphi(x)$ for $x \notin A$, thus there isn't $x \in A$ such that $y \in \varphi(x)$, $y \notin \bigcup_{x \in A} \varphi(x) = \varphi(A)$, $y \in \mathbf{C}(\varphi(A))$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

在 $T = \Phi(S)$ 下，若 $y \notin \Phi(A)$，从满值域选 $x \in S$ 使 $y \in \Phi(x)$；该 x 不在 A，故 $y \in \Phi(S \setminus A)$。反向需不同输入的输出集不相交，否则同一输出可同时由 A 内外产生。

19 原课件第 19 页：对应的图

Correspondences

The graph of the correspondence φ is a subset of $S \times T$, namely

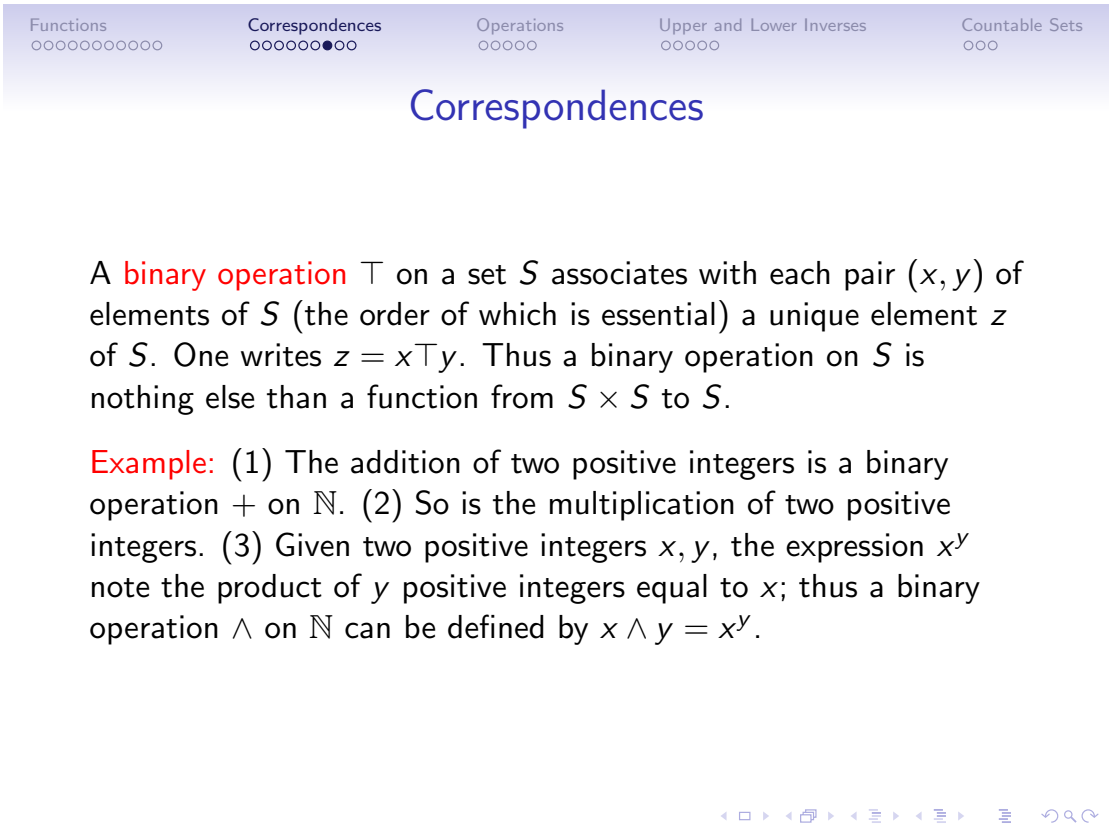
$$G(\varphi) = \{(x, y) \in S \times T \mid y \in \varphi(x)\}.$$

Example: Consider the case of the rectangle already used to illustrate the graph of a mapping: the graph of correspondence from S to T is visualized as a set of points of the rectangle: the intersection of this set with a vertical through an arbitrary point x of S is a non-empty set (the projection of which on T is $\varphi(x)$).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$G(\Phi) = \{(x, y) : y \in \Phi(x)\}$。竖直截面等于值集，可以有多个点。104中二元关系也能看成这类有序对集合。

20 原课件第 20 页：二元运算



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

运算是 $S \times S \to S$ 的函数，必须封闭且单值。正整数的减法不封闭，因此不是正整数上的二元运算；加法和乘法是。

21 原课件第 21 页：交换律与结合律

Correspondences

A binary operation $\top$ on S is said to be associative if for all x, y, z in S one has $(x \top y) \top z = x \top (y \top z)$. It is said to be commutative if for all x, y in S one has $x \top y = y \top x$.

Example: (1) Both $+$ and $\cdot$ on $\mathbb{N}$ are associative and commutative. (2) $\wedge$ on $\mathbb{N}$ is neither associative nor commutative.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

交换律交换两个数位置，结合律改变三个数的括号。幂运算不交换：$2^3 \neq 3^2$；不结合：$(2^2)^3 = 64$ 而 $2^{(2^3)} = 256$。一个反例足以否定全称命题。

22 原课件第 22 页：分配律

Correspondences

Consider two binary operations $\perp$ and $\top$ defined on S. The first is said to be distributive with respect to the second if for all x, y, z in S one has $x\perp(y\top z) = (x\perp y)\top(x\perp z)$.

Example: (1) On $\mathbb{N}$, $\cdot$ is distributive with respect to $+$. (2) But $+$ is not distributive with respect to $\cdot$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

乘法对加法分配：$a(b+c) = ab+ac$。加法不对乘法分配，例如 $1+(2\cdot 3) = 7$，但 $(1+2)(1+3) = 12$。运算顺序需逐层读取。

23 原课件第 23 页：对应逐点运算

Operations on mappings

If φ_1 and φ_2 are two mappings of S into T, their union is the mapping $(\varphi_1 \cup \varphi_2)$ of S into T defined by

$$(\varphi_1 \cup \varphi_2)(x) = \varphi_1(x) \cup \varphi_2(x).$$

The intersection of φ_1 and φ_2 is the mapping $(\varphi_1 \cap \varphi_2)$ of S into T defined by

$$(\varphi_1 \cap \varphi_2)(x) = \varphi_1(x) \cap \varphi_2(x).$$

The Cartesian product of φ_1 and φ_2 is the mapping $(\varphi_1 \times \varphi_2)$ of S into $T \times T$ defined by

$$(\varphi_1 \times \varphi_2)(x) = \varphi_1(x) \times \varphi_2(x).$$

More generally, given any operations $\vee$ and $\wedge$ on sets, we write

$$(\varphi_1 \vee \varphi_2)(x) = \varphi_1(x) \vee \varphi_2(x);$$
$$(\varphi_1 \wedge \varphi_2)(x) = \varphi_1(x) \wedge \varphi_2(x).$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

并、交、积均先固定输入 x 再对值集运算。交值可能为空，即使两个原值都非空，因此不能自动保留非空值条件。

24 原课件第 24 页：对应的复合

Operations on mappings

If φ_1 is a mapping of S into T and φ_2 is a mapping of T into U, the composition product of φ_2 by φ_1 is the mapping $(\varphi_2 \cdot \varphi_1)$ of S into U defined by

$$(\varphi_2 \cdot \varphi_1)(x) = \varphi_2(\varphi_1(x)).$$

We have

(1) $(\varphi_1 \cup \varphi_2)(A) = \varphi_1(A) \cup \varphi_2(A)$;
(2) $(\varphi_1 \cap \varphi_2)(A) \subseteq \varphi_1(A) \cap \varphi_2(A)$;
(3) $(\varphi_1 \times \varphi_2)(A) \subseteq \varphi_1(A) \times \varphi_2(A)$;
(4) $(\varphi_2 \cdot \varphi_1)(A) = \varphi_2(\varphi_1(A))$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$(\Psi \circ \Phi)(x) = \bigcup_{y \in \Phi(x)} \Psi(y)$，表示经某个中间点能到达的输出。逐点交的像一般小于像的交，因为后者可使用不同输入。积的像也可能只含同步产生的坐标对。

25 原课件第 25 页：常对应与恒等对应

Operations on mappings

A mapping Δ is said to be constant if there exists a subset C of T such that $\Delta(x) = C$ for all S. A constant mapping satisfies the following property:

$$(\Delta \cap \varphi)(A) = \Delta(A) \cap \varphi(A).$$

Another important mapping is the identity mapping, which is the single-valued mapping I of S onto S defined by

$$I(x) = \{x\}.$$

$$\begin{aligned}(\Delta \cap \varphi)(A) &= \bigcup_{x \in A} (\Delta \cap \varphi)(x) = \bigcup_{x \in A} [\Delta(x) \cap \varphi(x)] \\ &= \bigcup_{x \in A} [C \cap \varphi(x)] = C \cap \left[\bigcup_{x \in A} \varphi(x) \right] \\ &= C \cap \varphi(A) = \Delta(A) \cap \varphi(A).\end{aligned}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $\Delta(x) = C$，对非空 A 有 $\Delta(A) = C$，所以 $\bigcup_{x \in A}(C \cap \Phi(x)) = C \cap \Phi(A)$；空 A 时两边均为空。恒等对应的输出是单点集 $\{x\}$。

26 原课件第 26 页：复合结合与可达性

Operations on mappings

If $\varphi_1, \varphi_2, \varphi_3$ are mappings of S into S, we have associativity of composition product

$$\varphi_3 \cdot (\varphi_2 \cdot \varphi_1)(x) = (\varphi_3 \cdot \varphi_2) \cdot \varphi_1(x).$$

Thus, if φ is a mapping of S into S, we can write

$$\varphi^2(x) = \varphi(\varphi(x))$$
$$\varphi^3(x) = \varphi\left(\varphi^2(x)\right) = \varphi^2(\varphi(x)), \text{ etc.}$$

The transitive closure of φ is a mapping $\hat{\varphi}$ of S into S defined by

$$\hat{\varphi}(x) = \{x\} \cup \varphi(x) \cup \varphi^2(x) \cup \varphi^3(x) \cup \cdots$$

The correspondence $A \longrightarrow \hat{\varphi}(A)$ is a closure operation, for

(1) $\hat{\varphi}(A) \supseteq A$,

(2) $A \supseteq A' \Longrightarrow \hat{\varphi}(A) \supseteq \hat{\varphi}(A')$,

(3) $\hat{\varphi}(\hat{\varphi}(A)) = \hat{\varphi}(A)$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

复合结合来自交换“存在中间点”的分组。传递闭包 $\widehat{\Phi}$ 收集走零步、一步、两步等可达点。再走有限步仍为有限步，故 $\widehat{\Phi}(\widehat{\Phi}(A)) = \widehat{\Phi}(A)$；零步给扩张性，见证包含给单调性。

27 原课件第 27 页：半单值及单射

Operations on mappings

Theorem 7. If Γ_1 and Γ_2 are two semi-single-valued mappings, $\Gamma_1 \cap \Gamma_2$ and $\Gamma_1 \times \Gamma_2$ are semi-single-valued mappings.

Theorem 8. If one of the mappings Γ_1 and Γ_2 is injective, the mappings $\Gamma_1 \cap \Gamma_2$ and $\Gamma_1 \times \Gamma_2$ are injective.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

半单值指每个值集至多一个元素。交与积保留该性质，因为从每个因子至多各取一个元素。若某个因子在不同输入的值集不相交，则积中同一点不可能同时来自两个输入；交同理。

28 原课件第 28 页：逆对应

Upper and Lower Inverses

If φ is a mapping of S into T, its lower inverse is the mapping φ^- of T into S defined by

$$\varphi^-(y) = \{x \in S \mid y \in \varphi(x)\}.$$

This is a mapping whose domain is T^* and whose range is S^*.

Theorem 9. If φ is single-valued, φ^- is injective; if φ is injective, φ^- is single-valued; if φ is semi-single-valued, φ^- is semi-single-valued.

Proof. For $y \neq y'$ we want to prove $\varphi^-(y) \cap \varphi^-(y') = \varnothing$. Otherwise, there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is single-valued, then $y = \varphi(x) = y'$, which is a contradiction.

If $\varphi^-(y) \cap \varphi^-(y') \neq \varnothing$, then there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is injective, then $y = y'$ and hence $\varphi^-(y) = \varphi^-(y')$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\Phi^-(y) = \{x : y \in \Phi(x)\}$。若原对应每点恰一输出，则两个不同 y 的逆值不相交。若原对应不同输入值集不相交，则每个 y 至多有一个原像。纠错：证明后半必须使用“不同输入的值集不相交”，不能从单射推出同一输入有唯一输出。

29 原课件第 29 页：下逆像

Upper and Lower Inverses

For any non-empty subset B of T we have

$$\varphi^{-}(B) = \{x \in S \mid \varphi(x) \cap B \neq \varnothing\}$$

and we also write $\varphi^{-}(\varnothing) = \varnothing$. It is clear that the inverse of φ^{-} is $(\varphi^{-})^{-} = \varphi$ and that $y \in \varphi(x)$ is equivalent to $x \in \varphi^{-}(y)$.

$$\begin{aligned}
\left(\varphi^{-}\right)^{-}(A) &= \{y \in T \mid \varphi^{-}(y) \cap A \neq \varnothing\} \\
&= \{y \in T \mid \{x \in S \mid y \in \varphi(x)\} \cap A \neq \varnothing\} \\
&= \{y \in T \mid \{x \in A \mid y \in \varphi(x)\} \neq \varnothing\} \\
&= \{y \in T \mid y \in \varphi(A)\} = \varphi(A)
\end{aligned}$$

$$\left(\varphi^{-}\right)^{-}(x) = \{y \in T \mid x \in \varphi^{-}(y)\} = \{y \in T \mid y \in \varphi(x)\} = \varphi(x)$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$x \in \Phi^{-}(B) \iff \exists y \in B : y \in \Phi(x)$。交换有序对两坐标再交换回来不变，故逆对应的逆是原对应。集合逆像的“存在”含义对后面下半连续极重要。

30 原课件第 30 页：上逆像

Upper and Lower Inverses

Apart from the mapping φ^{-}, it is sometimes useful to consider another correspondence of a similar kind, which we call the upper inverse of φ; in this mapping to each subset B of T there corresponds the set

$$\varphi^{+}(B) = \{x \in S^{*} \mid \varphi(x) \subseteq B\}.$$

In particular, if $B = \varnothing$, we have $\varphi^{+}(\varnothing) = \varnothing$. We observe that for all subsets B we have $\varphi^{+}(B) \subseteq \varphi^{-}(B)$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\Phi^{+}(B) = \{x \in S^{*} : \Phi(x) \subseteq B\}$ 表示所有允许输出都在 B。非空值时“全部在内”推出“至少一个在内”，所以 $\Phi^{+}(B) \subseteq \Phi^{-}(B)$。

31 原课件第 31 页：补集对偶

Upper and Lower Inverses

Theorem 10. If B_1 and B_2 are subsets of T, then we have

(1) $\mathbf{C}\varphi^+(B) = \varphi^-(\mathbf{C}B)$ and $\mathbf{C}\varphi^-(B) = \varphi^+(\mathbf{C}B)$.

(2) $\varphi^+(B_1) \cup \varphi^+(B_2) \subseteq \varphi^+(B_1 \cup B_2)$.

(3) $\varphi^-(B_1) \cup \varphi^-(B_2) = \varphi^-(B_1 \cup B_2)$.

Proof.

$$\begin{aligned}\mathbf{C}\varphi^+(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \subseteq B\} \\ &= \mathbf{C}\{x \in S \mid \varphi(x) \cap \mathbf{C}B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap \mathbf{C}B \neq \varnothing\} = \varphi^-(\mathbf{C}B)\end{aligned}$$

$$\begin{aligned}\mathbf{C}\varphi^-(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \cap B \neq \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}B\} = \varphi^+(\mathbf{C}B)\end{aligned}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

在有效定义域 S^* 内，非“全部在 B”等价于“有一个在 $T \setminus B$”。因此 $S^* \setminus \Phi^+(B) = \Phi^-(T \setminus B)$。允许空值却取整个 S 作补集会破坏课件公式。

32 原课件第 32 页：并集的上下逆像

Upper and Lower Inverses

$$
\begin{aligned}
& \varphi^{+}(B_1) \cup \varphi^{+}(B_2) \\
= \; & \{x \in S \mid \varphi(x) \subseteq B_1\} \cup \{x \in S \mid \varphi(x) \subseteq B_2\} \\
\subseteq \; & \{x \in S \mid \varphi(x) \subseteq B_1 \cup B_2\} \\
= \; & \varphi^{+}(B_1 \cup B_2)
\end{aligned}
$$

$$
\begin{aligned}
& \varphi^{-}(B_1) \cup \varphi^{-}(B_2) \\
= \; & \{x \in S \mid \varphi(x) \cap B_1 \neq \varnothing\} \cup \{x \in S \mid \varphi(x) \cap B_2 \neq \varnothing\} \\
= \; & \{x \in S \mid (\varphi(x) \cap B_1) \cup (\varphi(x) \cap B_2) \neq \varnothing\} \\
= \; & \{x \in S \mid \varphi(x) \cap (B_1 \cup B_2) \neq \varnothing\} \\
= \; & \varphi^{-}(B_1 \cup B_2)
\end{aligned}
$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

下逆像保持并：有输出在 $B_1 \cup B_2$ 当且仅有输出在至少一个集合。上逆像仅有包含，因为所有输出可能分散在两集合。例值集 $\{0,1\}$，分别取 $B_1 = \{0\}, B_2 = \{1\}$ 即为严格包含。

33 原课件第 33 页：数列是函数

Countable Sets

A sequence $\left(x^1, x^2, \cdots, x^q, \cdots\right)$ of elements of a set S is an intuitive concept. Precisely, it is defined as a function from $\mathbb{N}$ to S. It will he denoted by (x^q).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

数列是 $\mathbb{N} \to S$ 的函数，x_n 是第 n 项，允许重复。集合不记录顺序及重复，数列记录；集合 $\{1,2\}$ 与交替数列 $1,2,1,2,\ldots$ 不能混为一谈。

34 原课件第 34 页：可数与有限乘积

Countable Sets

Intuitively, a set S is countable if it has at most as many elements as $\mathbb{N}$. In a precise fashion, a set S is defined as countable if it can be put in one-to-one correspondence with a subset of $\mathbb{N}$. When the countable set S is not empty one can always choose the corresponding subset of $\mathbb{N}$ so that it owns 1, and so that, whenever it owns two positive integers, it owns every positive integer between them. The image of $x \in S$ in the correspondence is then called the rank of x. Thus a set is countable if and only if all its elements can be ranked, no two different elements having the same rank. One can prove that

Theorem 11. The product of finite countable sets is countable.

Example: (1) $\mathbb{N} \times \mathbb{N}$ is countable. (2) However, one can prove that the set of subsets $\mathbb{N}$ of is not countable.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

按本课约定可数包括有限集。$\mathbb{N}^2$ 按坐标和依次列出：$(1,1),(1,2),(2,1),\ldots$，每个数对有限步后出现。归纳得有限多个可数集的积可数；这不意味着可数无穷积也可数。

35 原课件第 35 页：Cantor 对角证明

Countable Sets

Theorem 12. The set of subsets $\mathbb{N}$ of is not countable.

Proof: We show that there is no surjection from a set A to its power set 2^A. Consider any function $f : A \to 2^A$ and let

$$B = \{a \in A \mid a \notin f(a)\}$$

We claim that there is no $b \in A$ such that $f(b) = B$. Indeed, assume $f(b) = B$ for some $b \in A$. Then either $b \in B$ hence $b \notin f(b)$ which is a contradiction, or $b \notin B$ implying that $b \in B$ which is again a contradiction. Hence the map f is not surjective as claimed. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

假设 $f : A \to \mathcal{P}(A)$ 满射。定义 $B = \{a \in A : a \notin f(a)\}$，由满射存在 b 使 $f(b) = B$。于是 $b \in B \iff b \notin B$，两种可能都矛盾。关键是 B 依赖于假设中的整张列表，必与每个列出集合在其对应位置不同。